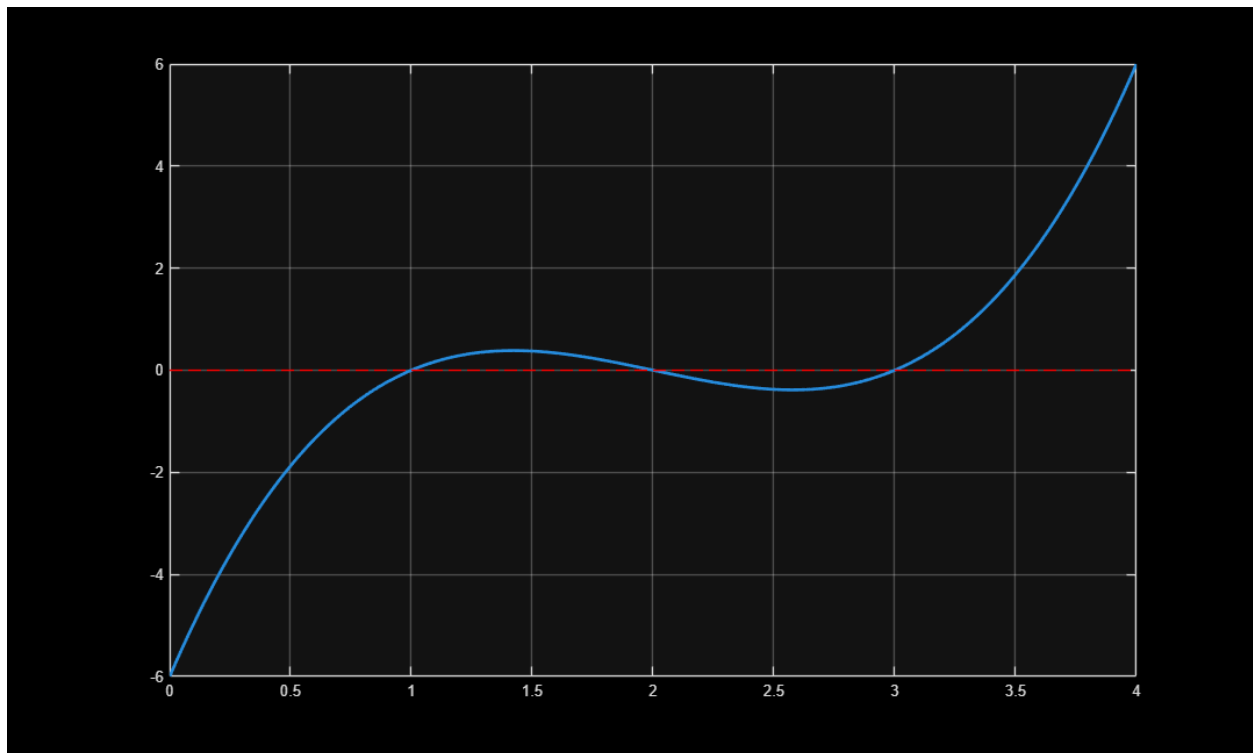

```

clc; clear; close all;
% Inclass_2a_1
% Newton-Raphson Method with one guessing

% Example:
f = @(x) x.^3 - 6*x.^2 + 11*x - 6;
f_p = @(x) 3*x.^2 - 12*x + 11; %First order deriv
%
x = linspace(0,4, 200);
y = f(x);
%
plot(x,y, '-', 'Linewidth', 2)
grid on
hold on
%
% add a red line at f = 0
plot([0 4], [0 0], 'r--')

```



Newton Raphson Method

```

xr = 3.2;
xr_old = xr;
iter = 0;
%
fprintf('    iter    xr        err \n')

while (1)

```

```

%Calculate f(xr) and f_p(xr)

FR = f(xr);
Fp = f_p(xr);

iter = iter + 1;

%Algorithm: Update the new guess

xr = xr - FR/Fp;

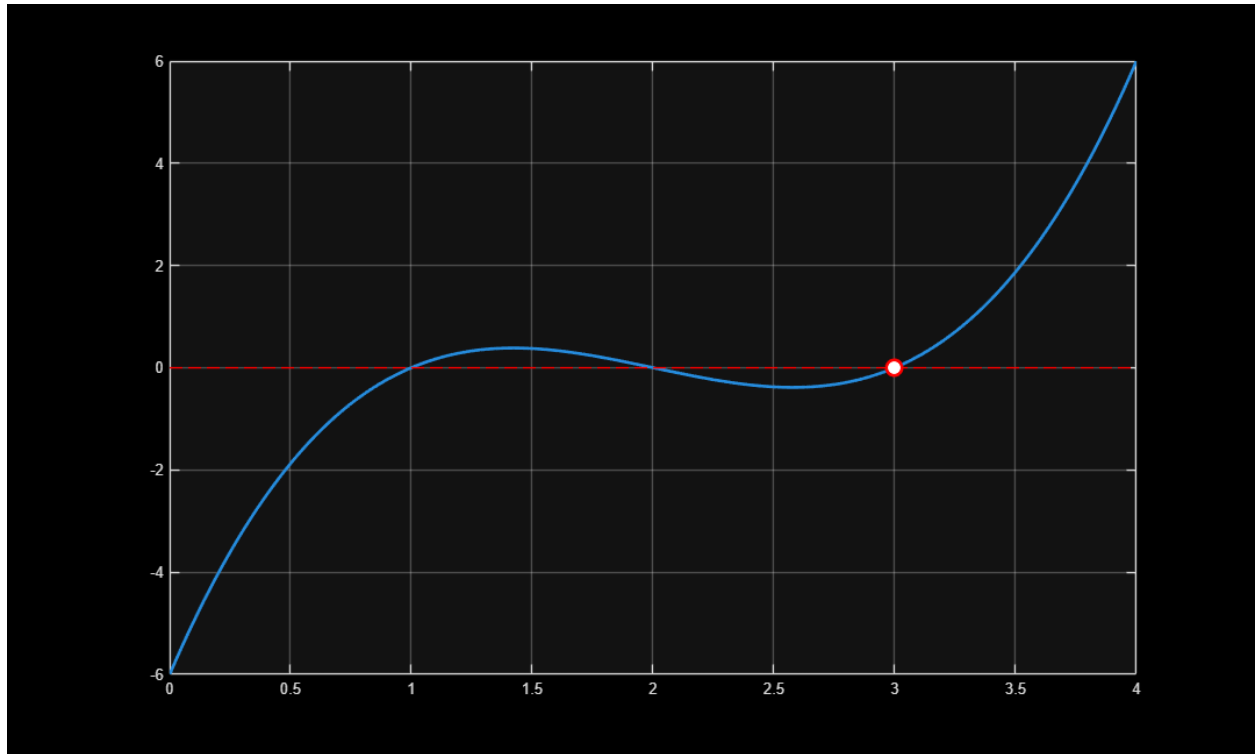
% calculate the relative error
err = abs((xr - xr_old)/xr);
%

fprintf('%5d %10.6f %10.5f \n',iter, xr, err)

if err < 1e-4 || iter > 20, break, end
%
xr_old = xr;
end
plot(xr, FR, 'o','LineWidth',2,...
      'MarkerEdgeColor','r',...
      'MarkerFaceColor','w',...
      'MarkerSize',10)

iter    xr          err
1    3.040964    0.05230
2    3.002298    0.01288
3    3.000008    0.00076
4    3.000000    0.00000

```



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